



Estd:1997

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# ST. ANN'S COLLEGE FOR WOMEN

Run by The Society of St Anne

Affiliated to Acharya Nagarjuna University, Approved by AICTE

Recognised under Section 2(f) of the UGC Act 1956, New Delhi.

Accredited by NAAC with 'A' Grade in the first cycle

**Amaravathi Road, Gorantla, Guntur-34, Andhra Pradesh, India.**



Website: [www.stannscollegeforwomen.org](http://www.stannscollegeforwomen.org)

## B.Sc Honours – Mathematics

### Empowering Students with Analytical & Logical Excellence

The **B.Sc Honours – Mathematics** programme at St. Ann's College for Women is a four-year undergraduate programme designed to build strong foundations in pure and applied mathematics, analytical reasoning, computational techniques, and problem-solving skills.

The programme focuses on mathematical structures, numerical analysis, algebra, calculus, statistics, and modern computational applications. It prepares students for careers in analytics, software development, banking, finance, research, education, and competitive examinations.

Students are trained through a combination of theoretical learning, practical applications, computational tools, and interdisciplinary minor courses that enhance employability and higher education opportunities.

**Note: St. Ann's College for Women presently offers the B.Sc Honours -Mathematics programme as a Three-Year Degree Programme in accordance with the Acharya Nagarjuna university and government guidelines**

### Programme Highlights

- Strong foundation in Pure & Applied Mathematics
- Training in Analytical & Logical Problem Solving
- Industry-oriented Computational Mathematics
- Practical Exposure through Programming & Mathematical Software
- Skill Development in Python, MATLAB & Data Analysis
- Research-oriented Curriculum
- Preparation for Banking, Finance & Competitive Exams
- Opportunities for Higher Education & Research

### Eligibility

Students seeking admission into the **B.Sc Honours – Mathematics** programme should fulfil the following criteria:

- Must have passed Intermediate / 10+2 from a recognized board

- Mathematics should be one of the subjects at Intermediate level
- Eligible Streams: **MPC, MEC, MBiPC**

### **Core Subjects Offered**

The programme includes advanced and application-oriented mathematics courses such as:

- Differential Equations
- Solid Geometry
- Group Theory
- Elementary Real Analysis
- Ring Theory
- Advanced Real Analysis
- Theory of Matrices
- Linear Algebra
- Vector Calculus
- Linear Programming
- Special Functions
- Numerical Methods
- Laplace Transforms
- Integral Transforms
- Advanced Numerical Methods
- Graph Theory
- Complex Analysis
- Mathematical Computations using Python
- Mathematical Methods using MATLAB
- Statistical Analysis using R

### **Proposed Minor Courses – Computer Science**

To enhance interdisciplinary knowledge and industry readiness, students can opt for minor courses in Computer Science including:

- Computer Fundamentals and Office Automation

- Computer Fundamentals and Office Automation – Practical
- Problem Solving Using C
- Problem Solving Using C – Practical
- Database Management System
- Database Management System – Practical
- OOPS Through JAVA
- OOPS Through JAVA – Practical
- Python Programming
- Python Programming – Practical
- Web Interface Design Technologies
- Web Interface Design Technologies – Practical

### **Career Opportunities**

Graduates of B.Sc Honours – Mathematics can explore career opportunities in diverse sectors such as:

- Data Analytics
- Banking & Finance
- Insurance
- Risk Management
- Software Development
- Research & Development
- Teaching & Academia
- Scientific Organizations
- Intelligence & Cryptography
- Statistical Analysis
- Technical Institutions

### **Job Roles After Graduation**

| <b>Job Position</b>  | <b>Description</b>                                                       |
|----------------------|--------------------------------------------------------------------------|
| Mathematician        | Applies mathematical theories and models to solve real-world problems    |
| Data Analyst         | Interprets and analyzes data to support business decisions               |
| Numerical Analyst    | Uses numerical methods for scientific and financial problem solving      |
| Cryptanalyst         | Develops and analyzes algorithms related to data security and encryption |
| Statistician         | Analyzes data and prepares statistical reports                           |
| Banking Professional | Works in banking, insurance, and financial sectors                       |

### **Higher Education Opportunities**

After completing the programme, students may pursue:

- M.Sc Mathematics
- M.Sc Computer Science
- MCA
- MBA
- Data Science & Analytics Programmes
- Research Programmes & Competitive Examinations

### **Scope of the Programme**

The demand for Mathematics graduates is continuously increasing across industries due to their analytical and computational skills. Graduates can build successful careers in:

- Research Organizations
- Universities & Academia
- Banks & Financial Institutions
- Insurance Companies
- Software Development Companies
- Data Analytics Firms

- Government Organizations
- Scientific Institutes
- Technical Institutions

The programme provides an excellent foundation for students aspiring for careers in technology, analytics, finance, education, and research.

### **Why Choose St. Ann's College for Women for B.Sc Honours – Mathematics?**

- Experienced and dedicated faculty with strong academic expertise
- Student-focused teaching methodology with personalized mentoring
- Well-equipped computer and mathematics laboratories
- Practical training in Python, MATLAB, R Programming & Computational Mathematics
- Industry-oriented curriculum aligned with emerging technologies
- Strong foundation for competitive examinations and higher education
- Skill enhancement through seminars, workshops, projects, and guest lectures
- Career guidance, placement support, and internship opportunities
- Safe, disciplined, and encouraging campus environment for women
- Holistic development through academics, research, co-curricular, and extracurricular activities
- Opportunities for interdisciplinary learning through Computer Science minor courses
- Focus on analytical thinking, logical reasoning, and problem-solving skills essential for modern careers

The programme at St. Ann's College for Women aims to empower students with mathematical excellence, technological skills, and professional confidence to succeed in academics, industry, and research.

### **Contact Us**

#### **St. Ann's College for Women**

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